

AI-driven short-term load forecasting enhanced by clustering in multi-type university buildings: Insights across building types and pandemic phases

Yu-Shin Hu^a, Kai-Yun Lo^a, I-Yun Lisa Hsieh^{a,b,*}

^a Department of Civil Engineering, National Taiwan University, No. 1, Sec. 4, Roosevelt Rd., Taipei, 10617, Taiwan

^b Department of Chemical Engineering, National Taiwan University, No. 1, Sec. 4, Roosevelt Rd., Taipei, 10617, Taiwan

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ABSTRACT

Accurate forecasting of electricity demand is pivotal for optimizing energy management in smart buildings and propelling the advancement towards net-zero goals. However, scalable, accurate, and robust building load prediction models are scarce in educational campuses, whose diverse functions and consumption patterns, resembling large-scale urban environments, are often hindered by data scarcity and lack of cross-building heterogeneity analysis. This study presents an AI-based short-term building load forecasting framework integrating K-means clustering with BiLSTM regression, tailored for various energy consumption patterns across different building types. The model requires no auxiliary variables and is adaptable to both regular operational conditions and disruptions caused by the COVID-19 pandemic. The clustering-enhanced model adeptly identifies unique energy consumption patterns and significantly improves prediction accuracy, with a 3.65 % increase in mean R^2 and a 55.19 % reduction in standard deviation under normal conditions. During the pandemic, its performance is further amplified, with a 4.90 % increase in mean R^2 and a 62.41 % reduction in standard deviation, highlighting its robustness. The model shows particularly high accuracy in buildings with consistent energy profiles, such as teaching and research facilities, while it encounters greater challenges in mixed-use and office buildings due to their variable energy patterns. The pandemic underscores the model's limitations in adapting to abrupt operational shifts, signaling a pressing need for future enhancements to incorporate adaptive forecasting techniques. This research substantiates the application of AI in building energy systems, contributing to the development of nearly zero-energy buildings (NZEB) and supporting the transition towards sustainable urban energy systems.

1. Introduction

1.1. Building energy challenges and the importance of prediction models

Buildings stand at the forefront of the global energy challenge, representing substantial contributors to both energy consumption and CO₂ emissions, accounting for 34 % and 37 %, respectively [1]. With forecasts predicting considerable growth in both electricity

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* Corresponding author. Department of Civil Engineering, National Taiwan University, No. 1, Sec. 4, Roosevelt Rd., Taipei, 10617, Taiwan.

E-mail address: iyhsieh@ntu.edu.tw (I.-Y.L. Hsieh).

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demand and built areas [2,3], the urgency for effective energy-saving measures is accentuated by global climate commitments, notably the Paris Agreement's goal to limit warming to 1.5 °C [4]. This target necessitates a 33 % reduction in building energy use by 2050 [5]. Educational facilities, which account for a significant portion of energy use—with electricity comprising 93 % of this consumption [6]—are notably under-researched in terms of energy conservation efforts, despite their substantial energy footprint [7]. The COVID-19 pandemic has further complicated energy management, as shifts to remote operations and altered building utilization patterns have disrupted traditional consumption norms [8].

The COVID-19 pandemic, declared a global pandemic by the World Health Organization on March 11th, 2020 [9], has further complicated energy management, with effects predominantly noticeable during periods of stricter lockdown measures [10]. Educational and research facilities were notably dominant in the energy consumption clusters showing increases and decreases. Specifically, educational buildings were frequently included in the decreased energy cluster, while research facilities were predominantly found in the increased energy cluster [11]. This finding aligns with the observation that changes in building load profiles during lockdowns are highly dependent on the building's purpose, with underutilized campus spaces contributing to significant energy wastage [12]. Furthermore, while lockdowns accelerated the shift to telework which is expected to remain a lasting trend [13], it may not be a sustainable mode of working or studying when compared to traditional commuting to centralized buildings, especially when the scope is expanded to include home office energy use [14].

In Taiwan, ambitious targets have been established to guide buildings towards near-zero carbon emissions, with a vision of over 85 % of existing buildings achieving near-zero carbon status by 2050 [15,16]. The median Energy Use Intensity (EUI) benchmarks managed by Taiwan's Executive Yuan categorize institutional buildings for comparative purposes [17,18]. Additionally, stringent EUI benchmarks have been specifically tailored for educational facilities [19]. Among Taiwanese universities, energy consumption averages at 79 kWh/m²/yr, rising to 114 kWh/m²/yr in top-tier universities with expansive teaching and research facilities—significantly higher than their counterparts [20]. Post-benchmarking, buildings are classified into different energy efficiency categories based on these standards. Yet, achieving these benchmarks remains a persistent challenge both in Taiwan and beyond. Studies in other regions have shown that a substantial proportion of buildings (62.2 %) exceed their benchmarked EUIs [21]. Similarly, the lack of detailed energy usage data available to stakeholders further complicates efforts to establish effective energy efficiency measures [22]. These limitations underscore the need for more effective energy prediction and benchmarking tools to bridge these gaps [23,24], a need further emphasized by recent electricity rate hikes in Taiwan [25].

Accurate prediction models for building energy are crucial for optimizing energy management and achieving carbon reduction and efficiency goals [26], while also playing a key role in smart grid applications [27]. These models facilitate effective monitoring, benchmarking, and evaluation of energy performance [28], supporting consumption management and the identification of savings opportunities [29]. They also critically inform the impacts of retrofits and renewable energy integrations, supporting data-driven policy-making [30]. Predominantly, energy prediction models are categorized into white-box, black-box, and grey-box models, based on their methodological frameworks: white-box models utilize physical equations; black-box models employ data-driven algorithms; and grey-box models integrate both approaches [31]. Notably, black-box models are recognized for their precision in addressing complex energy problems without the exhaustive requirements for detailed physical modeling [32].

1.2. Review and gaps of AI-based building load prediction models

Extensive literature has critically evaluated the application of data-driven models across various building contexts, including educational facilities, assessing their feature extraction, predictive modeling accuracy, and scope [33,34]. These studies also explore the influence of building types, characteristics, predicted energy types, and evaluation performance measures [35,36]. The potential of advanced machine learning algorithms, including deep learning (DL), is emphasized for their ability to handle large datasets, extract features, and model complex nonlinear phenomena [33]. Additionally, while some studies have employed time series decomposition methods to address the non-linear and nonstationary nature of building energy time series, this approach typically involves signal processing techniques followed by individual model construction and aggregation [37–39].

Energy forecasting spans a broad spectrum, from single-building analyses to multi-building and large-scale predictions. At the individual building level, studies employed IBWO-LSTM to forecast energy consumption in office buildings [40] and ANFIS in higher education buildings [41]. Other models, such as kCNN-LSTM and kLSTM, were used for energy consumption forecasting in an academic building, with K-means applied to time-series data for initial clustering based on monthly similarities [42,43]. Furthermore, the Energy Consumption Grouping (ECG) classification approach was utilized to cluster energy consumption data with similar daily patterns, thereby improving the predictive performance of ICEEMDAN-BiLSTM for a dormitory building [44].

In multi-building studies, research can generally be divided into single-type and multi-type building predictions. For single-type predictions, models such as TCN-LSTM with a selected-shared-private mechanism for Multi-Task Learning (MTL) were applied to model energy patterns in 14 residential buildings [45]. In addition, LSTM, GRU, and TFT were employed for both point and probabilistic predictions in two public historic buildings [46]. For cross-building type forecasting, buildings are often categorized based on their functional use to facilitate more targeted analyses [28,47,48]. Methods such as XGBoost, GRU, and LSTM were utilized for short-term cooling and heating load forecasting in district energy systems encompassing multiple buildings [49]. Moreover, a GWO-CNN-BiLSTM model was applied to predict daily, two-day, and weekly electricity consumption across a diverse range of building types, including campus, hospital, residential, and industrial buildings [50]. However, these studies largely overlooked the interdependencies among multi-building energy consumption patterns and often focused on a limited subset of buildings as representatives, potentially failing to capture inter-building influences and constraining the generalizability of their findings.

Another approach commonly used in multi-building studies is unsupervised clustering, which enhances predictive performance by

identifying similarities among time-series data and developing models based on correlated time series [51,52]. This approach eliminates the need for predefined data categorizations, which could introduce biases [53], thus enabling a fully data-driven framework. For instance, K-means combined with SVM was applied to predict energy consumption in 30 buildings by first grouping them into two clusters—one with relatively stable peak loads (e.g., educational, hospital, office, retail, and warehouse buildings) and another with more pronounced intra-day fluctuations (e.g., hotels, residential buildings, and restaurants)—and then using the cluster centroids as model inputs to represent typical energy consumption patterns [54]. However, this clustering method relies on predefined functional classifications, which do not account for the heterogeneity of energy consumption patterns within the same building category. Consequently, models developed based on functional classifications may struggle to capture more complex energy dynamics, either when buildings exhibit mixed usage and defy clear categorization, or when buildings within the same category show substantial variability in energy consumption.

On the other hand, in the context of COVID-19's impact on energy demand, several studies have employed data-driven methods to analyze and predict affected trends. TEDSE was used to forecast Turkey's electricity market, with the model's robustness and consistency being validated under both regular and pandemic conditions [55]. Methods such as ARIMA and KNN were applied to analyze energy price trends for both pre- and post-COVID-19 periods [56]. DSTNN was leveraged to predict demand distribution by directly linking demand patterns with spatiotemporal features, without relying on historical data, which improved long-term forecasting performance [57]. An auxiliary task for pandemic trend forecasting was also incorporated to support the main task by learning pandemic-related patterns. Moreover, a COVID-19-adjusted framework [58] used MLR to remove pandemic effects from electricity demand, with the residuals fed into LSTM-MSNet-DS for daily forecasting. This process helped remove patterns caused by the pandemic and lockdown measures, reducing computational complexity and improving prediction accuracy. However, such policy-based adjusted approaches may not be applicable for forecasts requiring higher temporal precision, such as hourly predictions, which are crucial for effective energy management. Additionally, while some studies incorporate training and testing data covering both normal and COVID-19 periods, they often lack epidemic-specific analyses and do not systematically evaluate the impact of unforeseen events, such as pandemics, on model accuracy and adaptability [44,46,50,59]. This highlights a gap in the assessment of model robustness for energy forecasting across varying operational periods.

Although the black-box models have been widely studied, the predictive capabilities for multiple buildings and diverse building types, particularly in educational settings, have received limited attention. Educational buildings, in particular, have been the subject of fewer in-depth, large-scale energy performance studies compared to commercial and residential buildings [52]. The significant variations in EUI among educational buildings [20], arising from their diverse characteristics and demands, are often overlooked, thus obscuring performance differences across individual buildings [45,52]. Furthermore, large-scale building energy performance forecasting is often dominated by aggregated-level predictions, neglecting inter-building relationships and performance disparities [60], and typically relies on annual predictions that lack the fine-grained temporal resolution essential for effective energy management [61]. Additionally, there is a lack of studies addressing model reliability under extreme and abnormal conditions, as well as epidemic-specific discussions. Other challenges regarding data availability and granularity remain critical for prediction accuracy [32,33], with data scarcity being especially pronounced in educational settings [62].

1.3. Research contribution and objectives

To bridge the identified research gaps, this study introduces an artificial intelligence (AI)-driven framework for short-term electricity load forecasting in educational campuses. Campus environments serve as ideal testbeds for energy forecasting models due to their diverse building functions and significant variations in energy consumption patterns, making them representative of large-scale, complex settings. Their transparent policy frameworks further facilitate comprehensive data collection for modeling and analysis. Additionally, unlike conventional models that often require extensive auxiliary inputs, this approach relies solely on historical EUI data, improving its applicability in data-scarce scenarios.

Beyond model development, this study conducts a comprehensive evaluation of the proposed framework across diverse spatial and temporal conditions, thoroughly considering variations in building types and the distinct phases of the COVID-19 pandemic. This evaluation ensures that the model not only captures spatial heterogeneity, which is often overlooked in large-scale forecasting, but also remains robust under evolving energy demand patterns. The findings further emphasize the need for adaptive and precise energy models that can effectively respond to fluctuating energy demand, especially during disruptive events such as the COVID-19 pandemic. The focus of the study includes the following key objectives:

- (1) Utilizing historical data to streamline model setup, reducing computational demands and setup time. This approach capitalizes on existing datasets to refine model accuracy and accelerate deployment, reflecting a smart integration of technology for sustainable energy management.
- (2) Analyzing the model's applicability across diverse building types, enhancing its scalability and generalizability. This adaptability is crucial in smart buildings, where different architectural and functional contexts require flexible and responsive energy solutions.
- (3) Improving the granularity of forecasting to match the dynamic nature of electricity demand more accurately. Enhanced forecasting supports efficient energy allocation and waste reduction, crucial for effective demand management strategies in intelligent energy systems.

- (4) Assessing the model’s reliability under extreme and non-normal conditions, ensuring consistent performance across unpredictable scenarios. Reliable energy forecasting strengthens the operational stability of smart buildings, supporting continuous energy optimization and occupant comfort even during adverse conditions.

These objectives collectively support the model’s accuracy and flexibility, providing a robust tool for managing energy that supports the transition to net-zero energy buildings within sustainable urban environments. By delivering dependable forecasting tools that perform robustly under variable conditions, this study contributes significantly to the advancement of smart energy management practices.

The remainder of this paper is organized as follows: Section 2 describes the methodology employed in this study, including data preparation, model development, and the framework for evaluating performance and conducting comparative analyses. Section 3 explores the determination process of the clustering model and analyzes the forecasting outcomes under various spatial and temporal scenarios. Finally, Section 4 concludes the paper by summarizing the main findings and contributions, discussing the limitations of the study, and proposing directions for future research.

2. Methodology and data

2.1. Research Framework

This study introduces a robust clustering-predictive model designed specifically for forecasting electricity load across various building types and temporal conditions. The methodological framework is outlined in three systematic steps, as illustrated in Fig. 1, guiding the construction and evaluation of building energy consumption predictions. Initially, the framework delineates the study boundaries and processes the data to ensure its cleanliness, consistency, and structure, thereby preparing it for effective modeling. Subsequently, the focus shifts to the construction and calibration of the proposed model, which involves clustering similar energy consumption patterns and developing personalized predictive models for each cluster. The final step involves using the proposed model to perform detailed performance evaluations and comparative analyses using real-world data.

2.2. Data preparation

2.2.1. Data representation and acquisition

In this study, we utilize hourly Energy Use Intensity (EUI), measured as energy consumption per square meter per hour (kWh/m²/h), as both the input and output variables. The input consists of historical EUI values for selected buildings, while the model predicts the EUI value for the next 1-h interval, providing a short-term forecast for the same buildings. This metric is chosen for its efficacy in reflecting building energy performance, serving as a robust benchmark for comparing energy efficiency across various buildings [63].

Our analysis focuses specifically on campus buildings at National Taiwan University (NTU), which provides a context-rich setting conducive to our research. Building floor area and load data, sourced from the NTU MAP [64] and the NTU Digital Electricity Meter Monitoring System [65], is processed to calculate the hourly EUI, offering a detailed perspective on energy usage patterns. The load data, recorded in 1-h intervals from January 2014 to December 2021, spans both normal operations and periods affected by the COVID-19 pandemic, which has influenced energy consumption patterns during these times.

2.2.2. Study boundaries delimitation

Our dataset includes 89 buildings located on the NTU campus, which provides a comprehensive snapshot of energy consumption across diverse functional types. Fig. 2 depicts the spatial distribution of these buildings, set within an urban campus environment. The choice of NTU as the study site is strategic, given the diversity of its building types and its substantial energy demand, making it a representative microcosm of densely populated metropolitan educational institutions. This setup enables a detailed analysis that is not

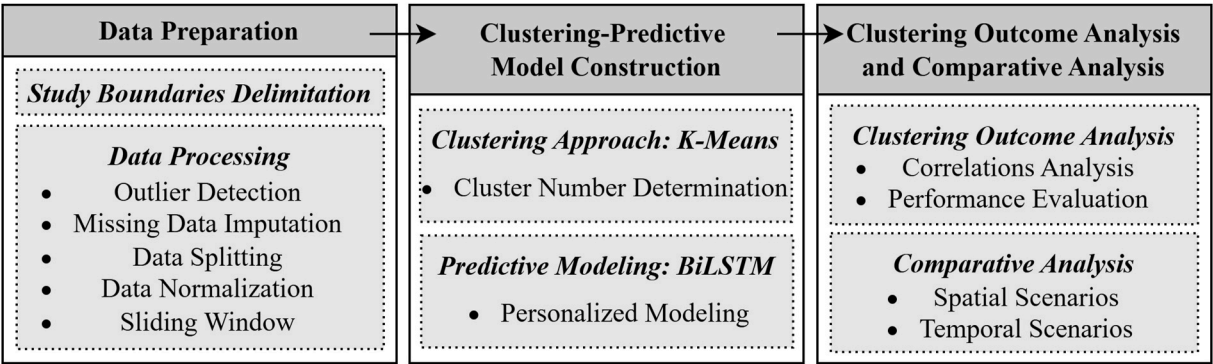


Fig. 1. Research framework.

only applicable to NTU but also offers empirical insights that can be generalized to other metropolitan educational settings.

2.2.3. Data processing

The data processing phase is crucial for refining the dataset associated with the selected buildings. This phase encompasses several essential steps designed to enhance the quality and reliability of the data for subsequent modeling. These steps include outlier detection, missing data imputation, data splitting, normalization, and the implementation of a sliding window technique.

2.2.3.1. Outlier detection. The data retrieved from the electricity meter monitoring system is subject to anomalies that could potentially distort the analysis. Consequently, initial preprocessing of the electricity meter data is essential to eliminate these irregularities and prevent potential bias in the model [66]. Common outlier detection methods include statistical, clustering-based, and distance-based approaches [67]. In this study, we evaluate four outlier detection techniques to ensure the integrity of our data analysis. The methods include the three-standard deviation (3-sigma) method and the Interquartile Range (IQR) method, known for their efficacy in identifying statistical outliers. Additionally, we implement the unsupervised Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm, which is particularly effective in discovering clusters and noise in spatial data [68]. To tailor our approach to the unique characteristics of each building's energy data, we also establish manually determined upper and lower thresholds customized for each dataset.

Fig. 3 presents a comparison of the four methods' efficacy in removing outliers from two randomly selected campus buildings—Computer and Information Networking Center and Global Change Research Center—from the dataset. The 3-sigma and IQR methods are found to be ineffective for handling complex time series data. While DBSCAN exhibits satisfactory performance, it is not fully effective in identifying all excessively low or high energy consumption values when compared to the manual threshold approach. Consequently, this study strategically adopts manually defined elimination thresholds for the outlier removal process to ensure precision and context-specific accuracy in our dataset. However, for scalability and automation in larger-scale systems, we specifically advocate for the use of the DBSCAN algorithm over traditional statistical techniques due to its superior capability to handle complex data structures efficiently. Moreover, to maintain the integrity of our modeling efforts, buildings with excessive amounts of missing data are excluded from the study to prevent biased or inaccurate outcomes. This rigorous data-cleaning process is critical for ensuring the reliability of our forecasting results.

2.2.3.2. Missing data imputation. Ensuring a complete dataset for input into our model is crucial, as gaps in data can significantly skew analytical outcomes. In many cases, missing values in datasets are considered to be missing completely at random (MCAR) [69], which simplifies the assumptions behind imputation methods. While previous studies have sometimes filled missing values using data from the preceding week [70], this method can become problematic if the initial weeks themselves contain numerous gaps. To address this

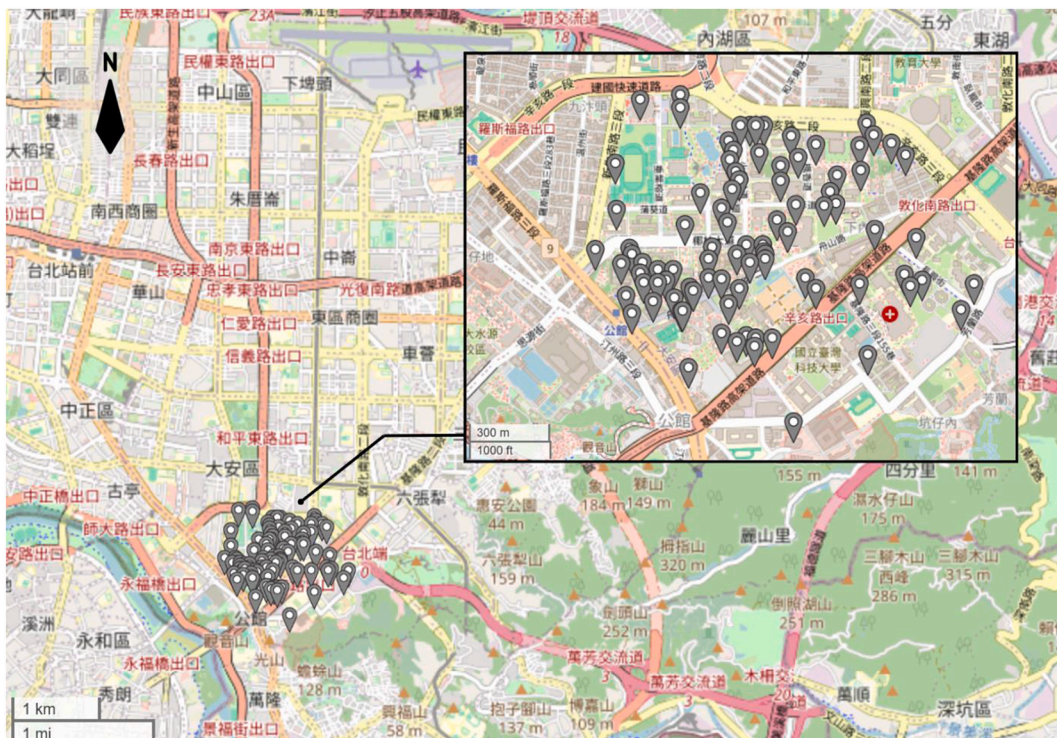
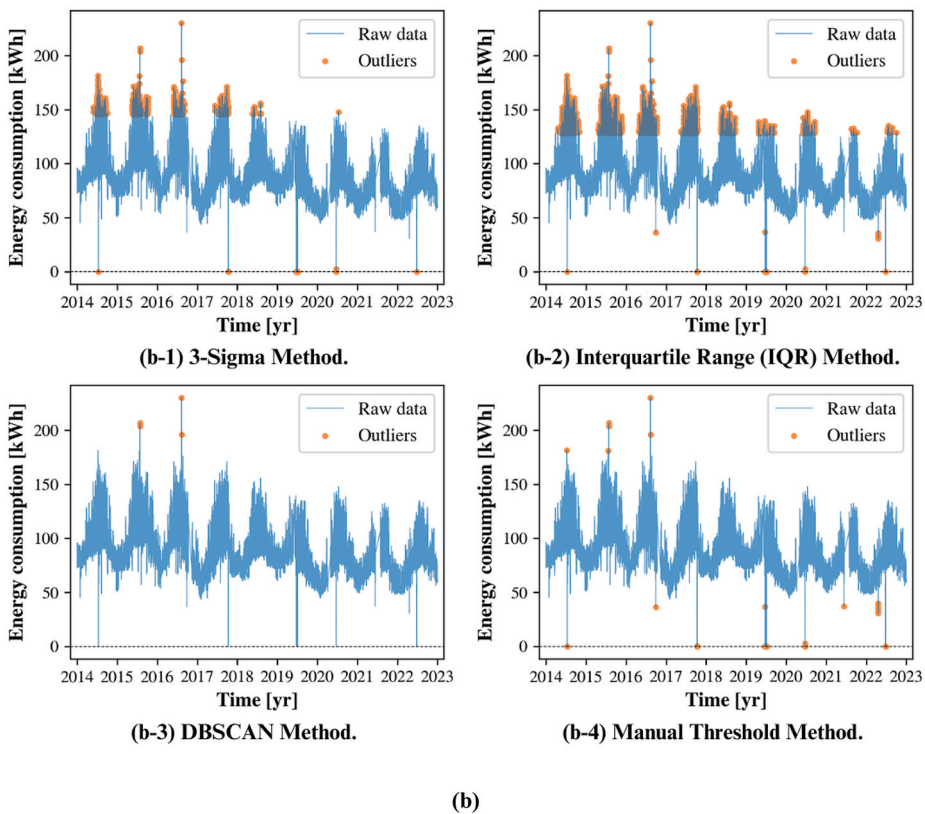
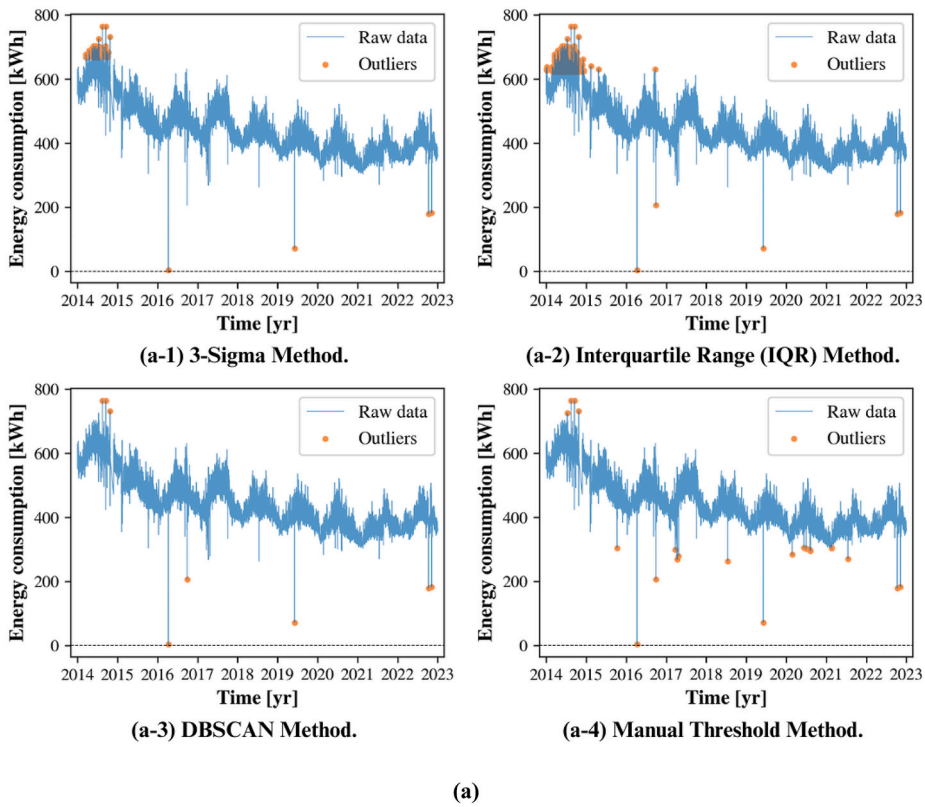


Fig. 2. Defined research boundary at NTU campus for urban building analysis.



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Fig. 3. Comparison of Outlier Detection Methods for Energy Consumption. This figure illustrates the effectiveness of various outlier detection methods with examples from (a) Computer and Information Networking Center and (b) Global Change Research Center, showcasing the methods of: (1) 3-Sigma, (2) Interquartile Range (IQR), (3) DBSCAN, and (4) Manual Threshold.

limitation more robustly, we employ KNNimputer [71], a non-parametric distance-based imputation method, commonly used in machine learning applications.

The KNNimputer leverages the k-nearest neighbors (KNN) algorithm, which identifies the closest neighbor samples using the Euclidean distance metric. This distance between two points x_i and x_j in our dataset is calculated as shown in Equation (1), where n is the dimension of the dataset and m is an index that runs through the dataset dimensions:

$$\text{dist}(x_i, x_j) = \sqrt{\sum_{m=1}^n (x_i^m - x_j^m)^2} \quad (1)$$

The imputation is then conducted by averaging the observed values from the nearest neighbors. This approach not only fills in the missing values but also helps maintain the statistical integrity of the dataset. The specific calculation for imputing a missing value x_m^* is based on the weighted average of these neighbors, detailed in Equation (2). Here, K denotes the number of nearest neighbors considered, and w_k is a weight that quantifies the similarity between the missing sample x_m^* and its k^{th} neighbor $x_{k,m}$:

$$x_m^* = \sum_{k=1}^K w_k x_{k,m} \quad (2)$$

2.2.3.3. Data splitting. Our data covers the period from January 1, 2014, to December 31, 2021, a timeframe that includes several distinct phases of building operation and external events such as the COVID-19 pandemic. To effectively model these varying conditions, we strategically segment the data:

1. **Training and Validation Phase:** Data from January 1, 2014, to December 31, 2018, are utilized to train and validate the predictive model. This period serves as a comprehensive baseline, allowing the model to learn typical fluctuations and trends in energy usage.
2. **Normal Testing Phase:** The subsequent year, from January 1, 2019, to December 31, 2019, is used as a test segment to evaluate the model under typical conditions, prior to any influence from the pandemic.
3. **Pandemic Assessment Phase:** The data from January 1, 2020, to December 31, 2021, covers the period affected by the COVID-19 pandemic. This segment is crucial for testing the model's resilience and adaptability to unprecedented and fluctuating conditions.

2.2.3.4. Data normalization. To maintain consistency across features of different scales and to facilitate more effective machine learning, we implement min-max normalization [72]. This technique adjusts each feature in the dataset to a common scale without distorting differences in the ranges of values. The normalization is governed by Equation (3), where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the original data, respectively. The normalized value x^* is then scaled to fall within new, predefined bounds $x_{\min}^*$ and $x_{\max}^*$:

$$x^* = \frac{x - x_{\min}}{x_{\max} - x_{\min}} (x_{\max}^* - x_{\min}^*) + x_{\min}^* \quad (3)$$

2.2.3.5. ACF and sliding window. The incorporation of historical data into energy consumption forecasting significantly enhances accuracy by utilizing the autocorrelation inherent in time series data [73]. One critical step in this process is identifying the optimal size for the input vector, which is crucial for effective model training. Such size should provide sufficient historical context to inform predictions, while avoiding the inclusion of irrelevant noise. To determine the most effective window size, we employ the Auto-correlation Function (ACF), a tool that measures the correlation between a current data point and its past values, or lags [74]. The ACF is calculated as the ratio of the covariance between the current value and its lagged values to the variance of the current value.

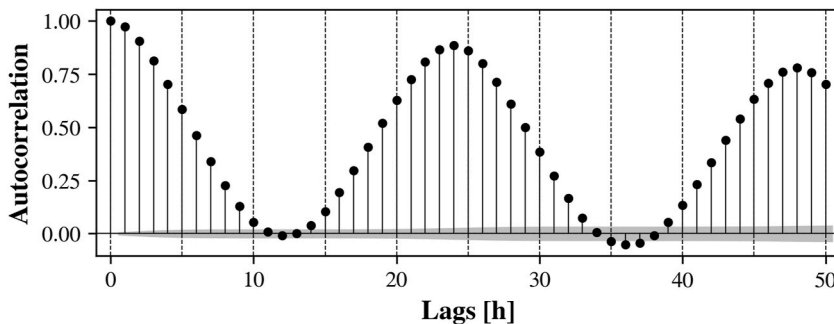


Fig. 4. Autocorrelation function analysis across lags.

Equation (4) illustrates this calculation, where x_t represents the time series value at time t , k is the lag, N is the total number of observations, and $\bar{x}$ is the mean of the time series:

$$ACF(k) = \frac{Cov(x_t, x_{t+k})}{Var(x_t)} = \frac{\sum_{t=1}^{N-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^N (x_t - \bar{x})^2} \quad (4)$$

The absolute value of the ACF indicates the strength of the correlation between the data points. A higher absolute value suggests a stronger correlation, which is crucial for selecting an effective window size. In addition, the significance of each lag is also assessed using a confidence interval. If the autocorrelation coefficient for a particular lag exceeds this interval, it is considered statistically significant. As demonstrated in Fig. 4, the ACF analysis shows that the autocorrelation at a 24-h lag reaches a peak, suggesting a strong correlation between the data point and its value 24 h prior. This result not only confirms the daily periodicity of electricity consumption but also underscores the statistical significance of this lag, as it exceeds the 95 % confidence interval.

To effectively utilize this correlated historical data for forecasting, we implement a sliding window mechanism with a time step of 1 h and a window size of 24 h. This setup allows us to use the preceding 24 h of data to forecast energy usage for the upcoming hour, enriching the model's input and enhancing the accuracy of our short-term, 1-h-ahead predictions. The process of training and predicting with this sliding window technique is depicted in Fig. 5.

2.3. Clustering-predictive model construction

2.3.1. Model architecture

In this research, we introduce a hybrid model, referred to as the clustering-predictive model in this study, that integrates K-means clustering with Bidirectional Long Short-Term Memory (BiLSTM) networks to forecast building energy consumption. Personalized BiLSTM models are developed for each cluster identified through data clustering techniques. Fig. 6 presents a schematic of the load forecasting framework, outlining its structural components and methodological workflow.

The architecture employs personalized predictive modeling, a specialized approach designed to create individualized prediction models for distinct groups or entities, each characterized by unique datasets and behavioral patterns [75]. This method contrasts with traditional models that apply a universal framework across all entities, often resulting in generalized and less accurate predictions. Moreover, this dual approach enhances forecasting accuracy by first grouping buildings with similar energy consumption patterns and then tailoring predictive models to each cluster [51], allowing for a more precise representation of the distinct characteristics inherent to each cluster. By adopting this methodology, our framework provides a more granular understanding of energy consumption patterns, addressing the complexities and variability inherent within each cluster.

The selection of clustering and predictive models is based on efficiency and performance. The K-means algorithm was chosen for its unsupervised approach, effectively clustering buildings by energy consumption patterns without predefined labels [52]. Its computational efficiency and ability to handle large, uniformly sized time series data make it well-suited for this framework [53]. For forecasting, BiLSTM was selected due to its ability to capture complex temporal dependencies crucial for energy prediction [76]. As highlighted in the literature review, LSTM is a widely established model, used independently, in variations, or in combination with other models. Additionally, a prior NTU campus study showed BiLSTM outperformed XGBoost, LSTM, GRU, and BiGRU in short-term energy forecasting [70]. Given these advantages and the research context, BiLSTM was deemed the most suitable model for this study.

2.3.2. Clustering approach: K-means

2.3.2.1. Overview of K-means algorithm. K-means clustering is utilized to group buildings based on similar energy consumption patterns over time. This unsupervised learning algorithm aims to partition the data into K distinct clusters by minimizing the intra-cluster variance, often measured as the Sum of Squares Error (SSE) [77]. This ensures that buildings within the same cluster exhibit closely related energy behaviors, enhancing the accuracy of the subsequent predictive model. The SSE is defined as Equation (5), where K is the number of clusters, C_i is the subset of data points in the i th cluster, and $dist(c_i, x)$ is the Euclidean distance between data points x and the cluster center c_i .

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist(c_i, x)^2 \quad (5)$$

The K-means algorithm begins by randomly assigning cluster centers. It then calculates the distances between these centers and

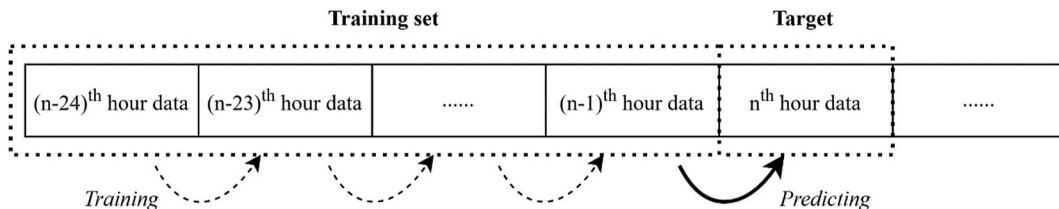


Fig. 5. Sliding window technique for training and forecasting.

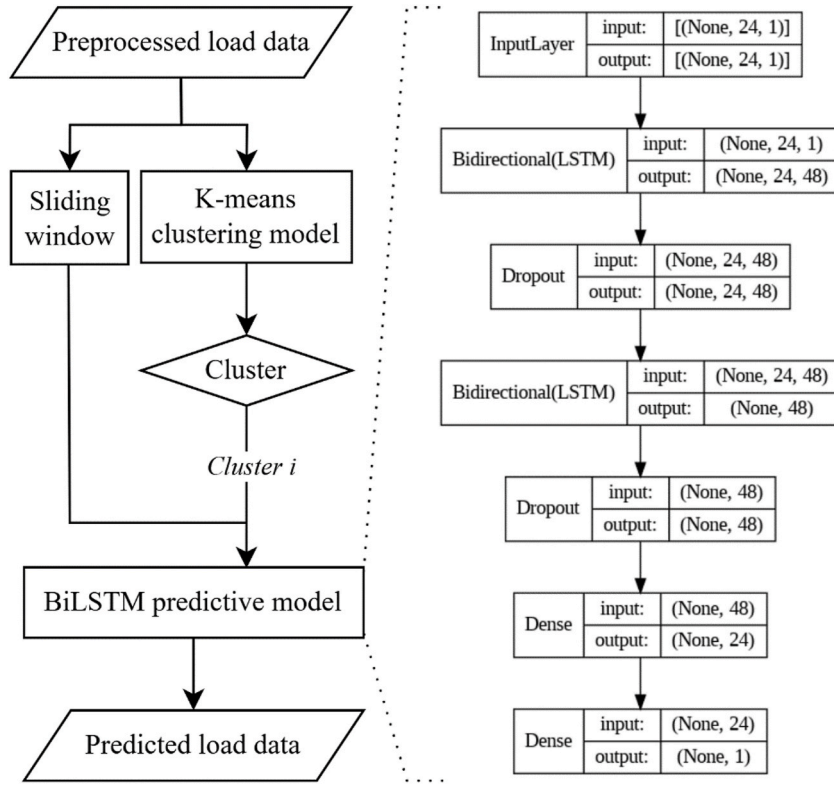


Fig. 6. Workflow of load forecasting framework using clustering-predictive model.

each data point, assigning each point to the cluster with the nearest center. Cluster centers are subsequently updated to be the mean of the points within each cluster. This process is repeated until convergence is achieved, indicated by minimal changes in the cluster centers and a minimized SSE. The optimization of SSE can be justified mathematically by calculating the derivative of SSE, demonstrating that setting the centers as the average values of the clusters effectively minimizes SSE.

2.3.2.2. Cluster number determination: Elbow Method and Silhouette Score Method. In the K-means algorithm, an increase in the number of clusters (K) typically leads to a corresponding decrease in SSE; however, this can result in overfitting by capturing noise rather than the underlying patterns. Therefore, identifying an appropriate K value that is less than the number of samples is crucial for balancing model complexity and generalizability in the K-means algorithm. We employ two widely recognized methods to determine the optimal K value: the Elbow method and the Silhouette Score method [78].

1. **Elbow Method:** The Elbow Method identifies the optimal K as the point where the Within-Cluster Sum of Squares (WCSS), also referred to as the SSE, sharply transitions from a rapid decrease to a more gradual decrease. This point, often called the “elbow,” represents a balance between cluster granularity and model simplicity. Here, increasing K further yields diminishing returns in terms of decreasing SSE, suggesting an optimal cluster count.
2. **Silhouette Score Method:** The Silhouette Score Method evaluates how well each data point fits within its cluster compared to other clusters by measuring both cohesion within the cluster and separation from other clusters. Scores range from -1 to $+1$, with higher values signifying better clustering quality. By evaluating the Silhouette Score, the optimal number of clusters K can be determined, ensuring both internal cohesion and external separation, which is essential for effective predictive modeling. The calculations are as Equations (6)–(9), where $d(x_i, x_j)$ is the distance between data points x_i and x_j .

$$\text{Intra-cluster distance : } a(x_i) = \frac{1}{|C_i| - 1} \sum_{j: x_j \in C_i, x_i \neq x_j} d(x_i, x_j) \quad (6)$$

$$\text{Nearest-cluster distance : } b(x_i) = \min_{k: C_k \neq C_i} \frac{1}{|C_k|} \sum_{j: x_j \in C_k} d(x_i, x_j) \quad (7)$$

$$\text{Silhouette score : } s(x_i) = \frac{b(x_i) - a(x_i)}{\min(a(x_i), b(x_i))} \quad (8)$$

$$\text{Average Silhouette Score (SC)} : SC = \bar{s}(x_i) = \text{avg} \left[\frac{b(x_i) - a(x_i)}{\min(a(x_i), b(x_i))} \right] \quad (9)$$

It involves two key components: the intra-cluster distance $a(x_i)$, which measures how close a point x_i is to others within the same cluster, and the nearest-cluster distance $b(x_i)$, which indicates how far x_i is from points in the nearest different cluster. A higher $b(x_i)$ and a lower $a(x_i)$ value indicate better clustering, as points are tightly packed within their own cluster and well-separated from others. The Silhouette Score $s(x_i)$ is then computed as the normalized difference between these two distances, and the Average Silhouette Score (SC) is the mean of these individual scores, assessing the overall clustering quality.

2.3.2.3. K-means configuration. To ensure the reproducibility of the clustering results, we provide the detailed configuration of the k-means clustering model. The number of clusters (K) is determined using the Elbow method and the Silhouette Score method, which assess the trade-off between model compactness and separation. The algorithm runs for a maximum of 300 iterations or until convergence, with a tolerance of $1e-4$. To reduce sensitivity to initialization, K independent runs with different initial centroids are performed, and the solution with the lowest inertia is selected.

2.3.3. Predictive modeling: BiLSTM

2.3.3.1. Overview of LSTM and BiLSTM networks. Bi-directional Long Short-Term Memory (BiLSTM) networks are utilized to model sequential dependencies in energy consumption data. Fig. 7 depicts the network architecture within a BiLSTM unit, highlighting the bidirectional data flow and how it integrates information from both past and future contexts; the expanded Long Short-Term Memory (LSTM) cell view shows gate interactions that control information flow and capture temporal dependencies.

BiLSTM extends the LSTM network, a specialized type of recurrent neural network (RNN) designed to address the limitations of traditional RNNs in capturing long-term dependencies. LSTM achieves this by incorporating a memory cell and a gating mechanism that regulates the flow of information, enabling effective learning of sequential patterns while mitigating issues like vanishing gradients. At each time step t , Equations (10)–(15) govern the forward propagation process of an LSTM unit, where h_t is the hidden state, x_t is the current input, and C_t denotes the cell state. Additionally, σ is the sigmoid activation function, W and b are the weight matrix and bias of network layers, and $\odot$ is the Hadamard product (element-wise product). The process involves four key stages: information discarding via the forget gate, which removes irrelevant data from the cell state; information addition through the input gate, which integrates new relevant inputs; cell state update, where the forget and input gate outputs modify the cell state; and output generation, where the output gate determines the final hidden state based on the updated cell state.

$$\text{Forget gate} : f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (10)$$

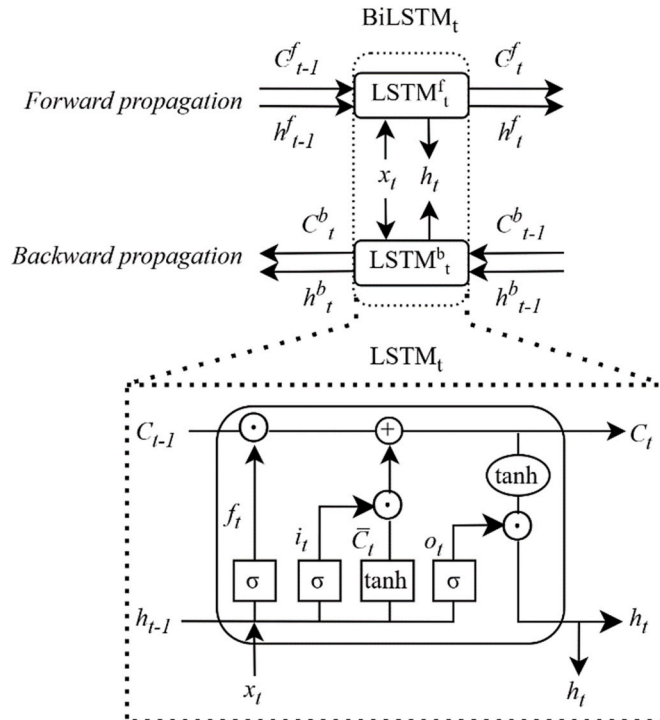


Fig. 7. BiLSTM unit structure with internal LSTM cell dynamics at time step t .

$$\text{Input gate : } i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (11)$$

$$\text{Candidate value : } \tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (12)$$

$$\text{Cell state : } C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (13)$$

$$\text{Output gate : } o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (14)$$

$$\text{Hidden state : } h_t = o_t \odot \tanh C_t \quad (15)$$

BiLSTM networks enhance the capabilities of traditional LSTMs by processing data in both forward and backward directions, thereby capturing both past and future contexts [79]. This bidirectional structure enhances the understanding of temporal patterns, making it particularly advantageous for complex forecasting tasks where context from both directions is important. At each time step, both the forward and backward LSTM units generate outputs as described in Equations (10)–(15). These outputs are then combined to produce the final hidden state, as shown in Equation (16), providing a more comprehensive understanding of the sequential data by considering both past and future contexts.

$$h_t = [h_t^f \oplus h_t^b] \quad (16)$$

2.3.3.2. BiLSTM configuration. To ensure the reproducibility and transparency of the BiLSTM model used in this study for capturing sequential dependencies in time-series data, we provide a detailed description of the model architecture, key hyperparameters, and training settings. As illustrated in Fig. 6, the configuration is as follows:

1. **Input Layer:** Incorporating a 24-h lagged EUI window, this layer captures the historical 24-h temporal dependencies based on the autocorrelation results in Fig. 4, establishing the foundation for subsequent analysis.
2. **BiLSTM Layers:** Two BiLSTM layers are employed to capture both forward and backward dependencies, allowing the model to account for temporal influences from both past and future states, thereby providing a more comprehensive understanding of energy patterns. Each layer has 24 hidden units, with tanh activation introducing non-linearity.
3. **Dropout Layers:** To mitigate overfitting, dropout layers are added after each BiLSTM layer with a rate of 15 %, improving generalizability by reducing feature reliance.
4. **Dense Layers:** The model ends with two dense layers: the first, with 24 units and a ReLU activation function, facilitates nonlinear transformations; the second layer, consisting of a single unit, performs a linear transformation to produce the final 1-h EUI forecast.
5. **Training Settings:** The model is trained with a batch size of 100 and a learning rate of 0.0001 to ensure stable learning. These parameters are selected to facilitate a gradual learning process and reduce the risk of divergence caused by excessively large updates. The Mean Squared Error (MSE) loss function is employed to minimize the discrepancy between predicted and actual values.

2.4. Performance evaluation and comparative analysis

2.4.1. Model performance evaluation metrics

To rigorously evaluate the performance of our predictive models, we employ several metrics that measure the accuracy and efficacy of the models during validation phases:

1. **Mean Absolute Error (MAE):** This metric quantifies the average magnitude of the errors in a set of predictions, without considering their direction. It's calculated as:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (17)$$

where y_i is the actual value and $\hat{y}_i$ is the predicted value at the i^{th} data point.

2. **Coefficient of Variation of Root Mean Square Error (CVRMSE):** This metric provides a normalized measure of the root mean square error, offering a perspective on error relative to the mean of the observed data:

$$CVRMSE = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}}{\bar{y}} \quad (18)$$

where $\bar{y}$ is the mean of the actual values, providing a sense of scale to the error.

3. **R-squared (R^2):** This coefficient of determination measures the proportion of the variance in the dependent variable that is predictable from the independent variables:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

(19)

By analyzing these metrics, we can ensure that the models are accurately predicting energy consumption and are well-suited for deployment in real-world energy management systems.

2.4.2. Comparative analysis

2.4.2.1. *Spatial analysis across building types.* Our study involves a detailed analysis of energy usage patterns across 89 buildings on the NTU campus, which are segmented into eight functional categories: Teaching (7), Research (49), Office (3), Mixed-use (8), Libraries (1), Sports centers (1), Dormitories (10), and Others (10). This classification, based on the primary functions of the buildings, allows for targeted studies into the specific energy consumption behaviors of each category, according to established methodologies that differentiate buildings by their functions for more precise energy studies [7,21,23,29]. Table 1 outlines the specific criteria used to categorize these buildings into distinct groups.

Fig. 8 illustrates the annual EUI distribution by building type, benchmarked against Taiwan’s Near-Zero Carbon Building Plan [15]. This analysis reveals that 46.07 % of the buildings on the NTU campus currently do not meet the target energy efficiency benchmark of EUI less than 100 kWh/m²/yr, highlighting the ongoing challenges and efforts required to enhance energy efficiency across campus facilities.

2.4.2.2. *Temporal analysis across epidemic phases.* The temporal analysis investigates the influence of the COVID-19 pandemic on energy consumption at NTU by segmenting the timeline into six distinct periods. Each phase corresponds to specific changes in campus protocols and remote learning practices, reflecting the university’s adaptive measures in response to the evolving pandemic situation:

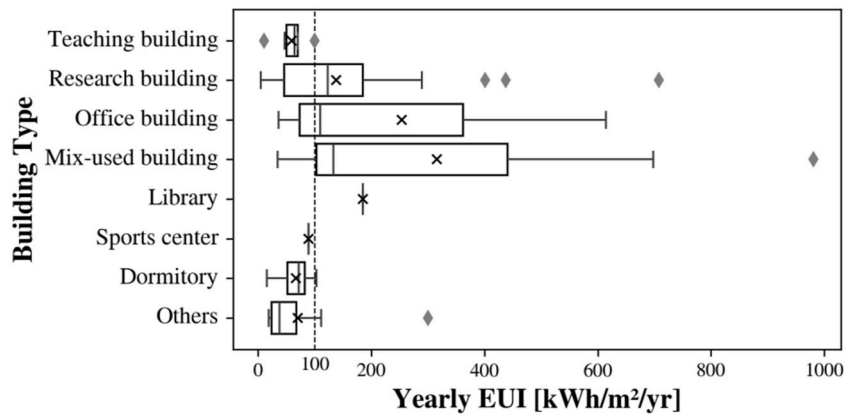
1. P1: Initial Phase - Minimal preventive measures were implemented during the early stages of the pandemic in Taiwan.
2. P2: Increased Remote Teaching - As the pandemic progressed, there was a significant shift towards remote teaching to reduce on-campus presence.
3. P3: Temporary Relaxation of Measures - There was a temporary easing of remote learning restrictions, allowing some flexibility in campus activities.
4. P4: Reinforcement of Stringent Measures - In response to rising case numbers, strict epidemic prevention measures were reinstated.
5. P5: Partial Relaxation - Following directives from the National Health Command Center, there was a slight easing of restrictions, allowing limited resumption of on-campus activities.
6. P6: Easing of In-Person Restrictions - Restrictions on in-person teaching activities were further relaxed, marking a gradual return towards normal campus operations.

Table 2 details the six distinct periods of COVID-19 at NTU, outlining the specific dates, descriptions of the pandemic’s progression, and the corresponding remote teaching regulations that were enforced. The timeline illustrates how the university’s operational responses adapted over the course of the pandemic to mitigate risks while maintaining educational activities.

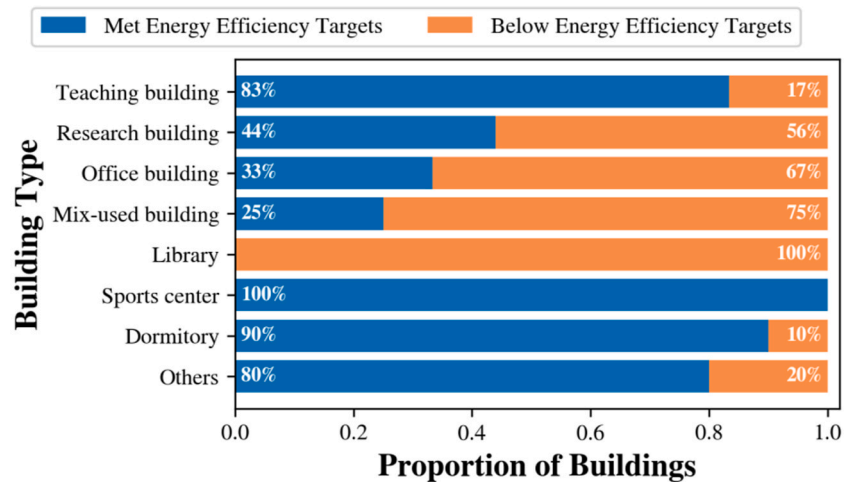
Fig. 9 tracks the cumulative changes in NTU’s energy consumption throughout the COVID-19 pandemic, contrasting the observed usage with the previous year’s data to highlight the impact of varying pandemic response phases. Initially, in phase P1, the energy usage nearly aligns with typical levels from preceding years, as minimal preventive measures were in place. However, with the onset of stricter remote teaching policies during phase P2, a noticeable reduction in energy consumption occurs. This trend of decreased usage slightly continues into P3, when measures were relaxed, suggesting a lingering influence of the initial restrictions. As NTU faced more severe pandemic conditions in P4, the graph shows a significant dip in energy usage, reflecting the comprehensive restrictions reimposed to curb the spread of the virus. Conversely, phase P5 sees a cautious increase in energy use, indicating a gradual move towards normalization following the partial lifting of restrictions by the National Health Command Center. The final phase, P6, exhibits a slight decrease in energy usage again, staying below normal levels as the university adapted to new operational norms with minor ongoing restrictions. This graphical representation underscores the adaptive challenges faced by NTU and its responses, which

Table 1
Classification rules of building types according to their function of use.

Building Category	Detailed Classification Rules
Teaching Building	Includes dedicated teaching facilities such as classrooms and departmental buildings without laboratories. Excludes buildings classified as ‘Research Buildings.’
Research Building	Comprises buildings that house laboratories or research centers, supporting scientific and technical inquiry.
Office Building	Encompasses administrative buildings, including offices and conference rooms that facilitate administrative functions.
Mixed-use Building	Buildings that serve multiple purposes, often including restaurants, shops, event centers, or health centers, providing a combination of services and amenities.
Library	Dedicated to library services, these buildings focus on study and the housing of academic resources.
Sports Center	Facilities specifically designed for sports activities and physical education.
Dormitory	Residential buildings that provide housing for students and faculty.
Others	Includes buildings with unique uses not covered in other categories, such as warehouses, guardhouses, and exhibition halls.



(a)



(b)

Fig. 8. EUI Analysis by Building Type at NTU: (a) Yearly EUI Distribution Marked Against Taiwan's Energy Efficiency Benchmark; (b) Proportion of Buildings Meeting vs. Not Meeting Taiwan's Energy Efficiency Goals Within Each Category. Note: The box plots illustrate the median (line), average (cross), and interquartile range (IQR), extending from the 25th to the 75th percentiles, with whiskers depicting the overall data range and diamond shapes indicating outliers.

significantly influenced campus energy dynamics during the pandemic.

3. Result and discussion

3.1. Cluster number determination and analysis

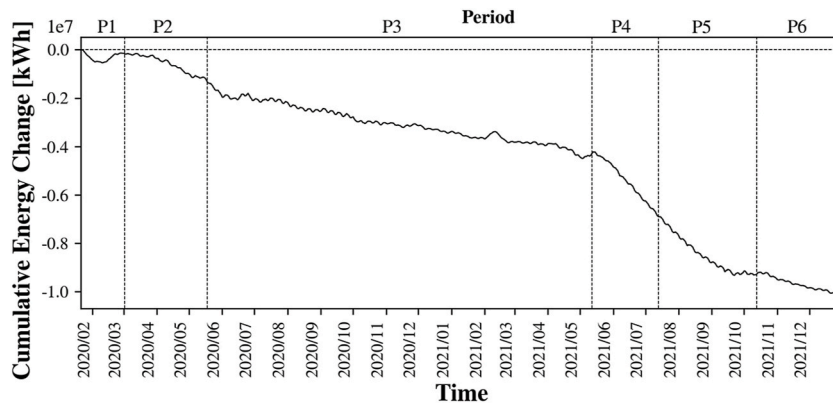
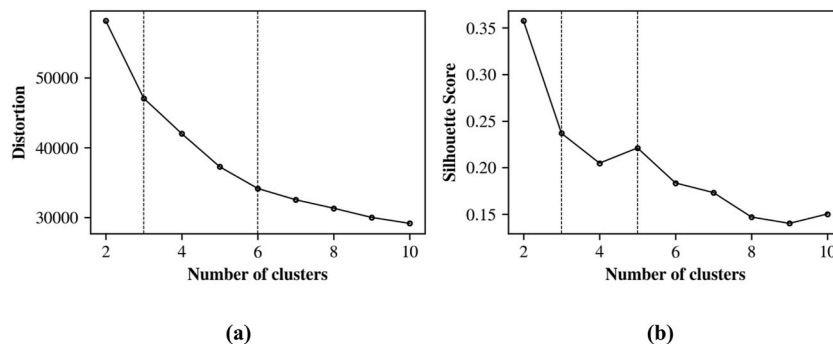
3.1.1. Optimal number of clusters

The process of determining the optimal number of clusters for categorizing buildings based on energy usage patterns is captured in Fig. 10. Using the Elbow method, the graph identifies distinct inflection points on the graph, suggesting potential optimal cluster numbers at $K = 3$ and $K = 6$. Conversely, the Silhouette score method, which evaluates the goodness of fit of the clustering, points to $K = 3$ and $K = 5$ as the most suitable choices based on higher scores. Balancing both analytical approaches, $K = 3$ is selected as the optimal number of clusters for further analysis, providing a clear balance between within-cluster homogeneity and between-cluster heterogeneity.

Table 2

Timeline and remote learning phases during COVID-19 at NTU.

Period	Dates	Description	Remote Teaching Details
P1	21/01/20-01/03/20	Initial stage of the pandemic with the first confirmed case in Taiwan.	–
P2	02/03/20-17/05/20	Delayed start of the semester due to escalating pandemic concerns.	Remote teaching from 13/04/20 to 26/04/20 for classes with more than 100 attendees. From 27/04/20 to 17/05/20 for classes with more than 60 attendees.
P3	18/05/20-11/05/21	Temporary relaxation of remote learning regulations.	Remote teaching from 18/05/20 to 01/06/20 for classes with more than 100 attendees.
P4	12/05/20-12/07/21	Enhanced epidemic prevention measures as pandemic conditions intensified.	Continuous remote teaching: 03/09/21 to 12/10/21 for all classes. 17/05/21 to 27/06/21 for all classes. 28/06/21 to 12/07/21 for classes with more than 50 people.
P5	13/07/21-12/10/21	Partial lifting of restrictions following guidance from the health authorities.	Remote teaching from 13/07/21 to 02/09/21 for classes with more than 50 attendees. From 03/09/21 to 12/10/21 for all classes.
P6	13/10/21-31/12/21	Further easing of measures to prioritize in-person teaching.	Remote teaching from 13/10/21 to 31/12/21 for classes with more than 80 attendees.

**Fig. 9.** Trends in cumulative energy change at NTU during COVID-19 relative to the previous year.**Fig. 10.** Optimal cluster determination: (a) Elbow and (b) Silhouette score analysis.

3.1.2. Temporal, spatial, and functional correlations of energy clusters

This subsection explores the temporal, spatial, and functional correlations of the derived energy usage clusters, with building functional categories as defined in Table 1. First, the functional correlation is examined in Fig. 11(a), which presents the distribution of building functional types within the three identified clusters, offering insights into the variability of energy consumption patterns across different types of buildings. The analysis reveals that buildings of the same functional type, such as research facilities or teaching spaces, are distributed across multiple clusters, indicating that the clustering captures variations in energy use behaviors rather than strictly grouping by function. This dispersion highlights the diverse energy use behaviors within each building type, raising questions about the efficacy of uniform benchmarking approaches for such classifications.

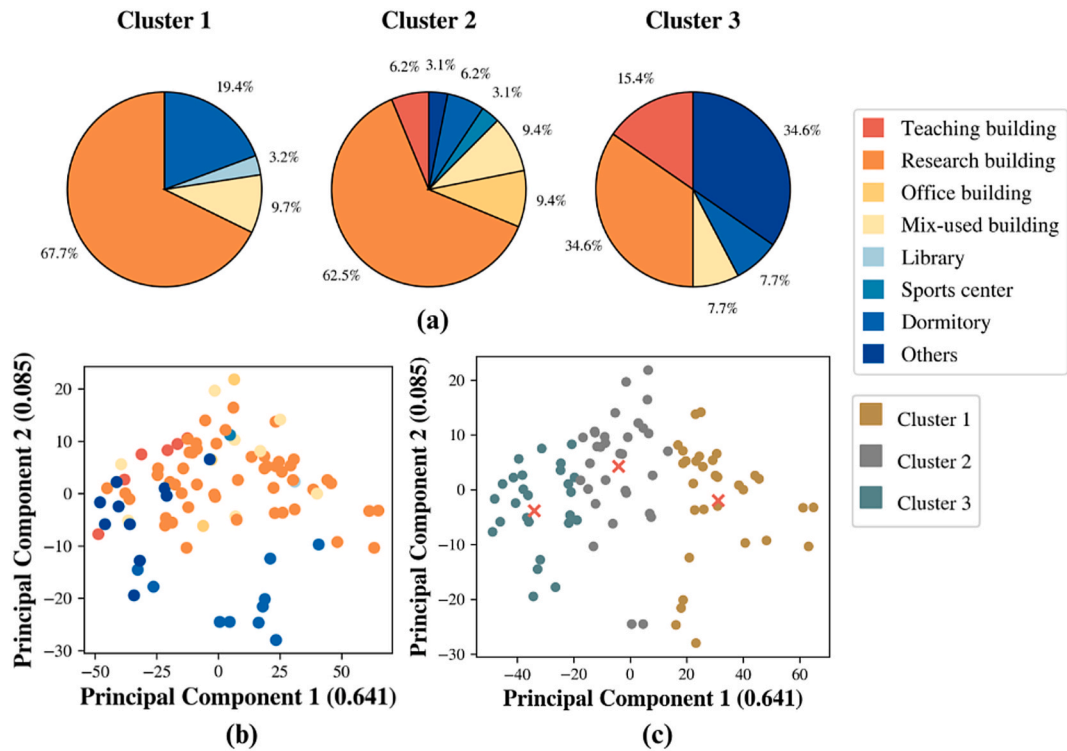


Fig. 11. Building Function and Energy Clusters: (a) Distribution of Building Functional Types in Identified Clusters; PCA of Normalized EUI Data (b) by Building Function Categories and (c) by K-means Clusters. Note: Principal components represent transformed axes derived from PCA that capture the most significant variance in the normalized EUI data, allowing visualization of similarities and differences among buildings.

Further analysis using Principal Component Analysis (PCA) is shown in Fig. 11(b) and (c), which visualizes the alignment between cluster assignments and building functionality categories. PCA reduces the dimensionality of the data by transforming it into a set of orthogonal principal components (PCs), which reveal dominant energy consumption patterns and capture the most significant variance in energy usage. The PCA plot underscores that while there is some grouping by building types, the overlap is considerable, indicating a weak correlation between the functional classification of buildings and their energy usage patterns. The combined findings from these analyses suggest that while clustering can reveal meaningful patterns in energy usage, the traditional method of categorizing buildings solely by their functional use may not be sufficient to guide effective energy management strategies [23]. This underscores the need for more tailored approaches that consider the specific energy dynamics and operational characteristics of each building category.

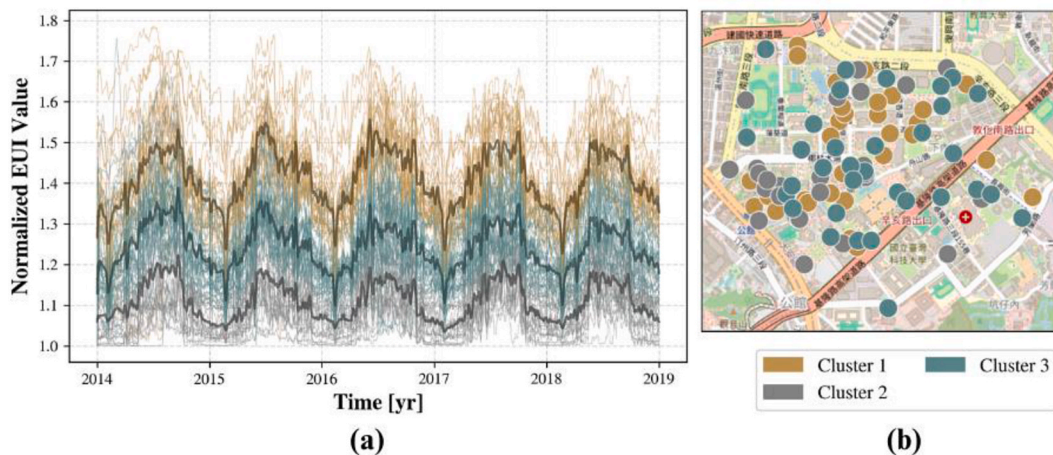


Fig. 12. Temporal and spatial analysis of energy clusters: (a) Temporal trends of normalized EUI, and (b) spatial distribution.

In addition, Fig. 12 examines the temporal and spatial analysis of energy clusters, highlighting distinct patterns in normalized EUI across buildings. The smoothed temporal trends (Fig. 12(a)) reveal strong seasonal variations across all clusters, with peaks in summer and lower values in winter, reflecting the influence of cooling demands. The campus schedule further shapes these trends, with a sharp decline in energy usage during the winter break and a slight reduction in peak demand during the summer break, indicating decreased occupancy and operational intensity. Cluster 1 demonstrates the highest EUI values, suggesting more intensive energy consumption and greater fluctuations, whereas Cluster 2 shows the lowest EUI values, reflecting relatively lower energy demands with smaller variations. Cluster 3 exhibits an intermediate pattern, reflecting a mix of energy-intensive and stable-use behaviors. The persistence of these trends across multiple years suggests that energy consumption patterns remain consistent, reinforcing the effectiveness of clustering in capturing distinct usage characteristics.

The spatial distribution (Fig. 12(b)) indicates that Clusters 1 and 3 are more concentrated in high-traffic areas of the campus, particularly in the central to upper-right region, suggesting higher energy consumption in these zones. In contrast, Cluster 2 buildings tend to be located outside these areas, potentially indicating lower overall energy demand or different operational characteristics. However, despite these localized patterns, the overall spatial distribution of energy clusters still exhibits a high degree of stochastic clustering, suggesting that geographic location alone does not determine energy usage behavior.

3.2. Model performance evaluation

The model was trained on a high-performance Linux machine with an NVIDIA GeForce RTX 4090 GPU, providing significant computational acceleration. The system's total available memory was 62.57 GB, and the training process involved training on approximately 100 buildings. Each complete training cycle for all buildings required approximately 400.38 s, with multiple iterations performed to optimize model performance.

3.2.1. Performance comparison of modeling approaches across conditions

The efficacy of the proposed clustering-predictive model is assessed by comparing it against a standard predictive-only model in scenarios typical of pre-pandemic and pandemic conditions. Fig. 13 shows the improvement in mean values and standard deviations for error metrics, including MAE, CVRMSE, and R^2 , achieved by the clustering-predictive model. The one standard deviation interval around the mean R^2 is truncated within the 0–1 range to avoid misinterpretation.

Under pre-pandemic conditions, the clustering-predictive model achieves a 1.42 % reduction in mean MAE, indicating that the incorporation of cluster-based adjustments effectively reduces the mean error relative to the predictive-only model. The standard deviation of MAE shows a slight 0.55 % increase, suggesting that the improvement in average accuracy is accompanied by stable variability. The mean CVRMSE decreases by 4.01 %, implying enhanced predictive stability and reduced error variability across

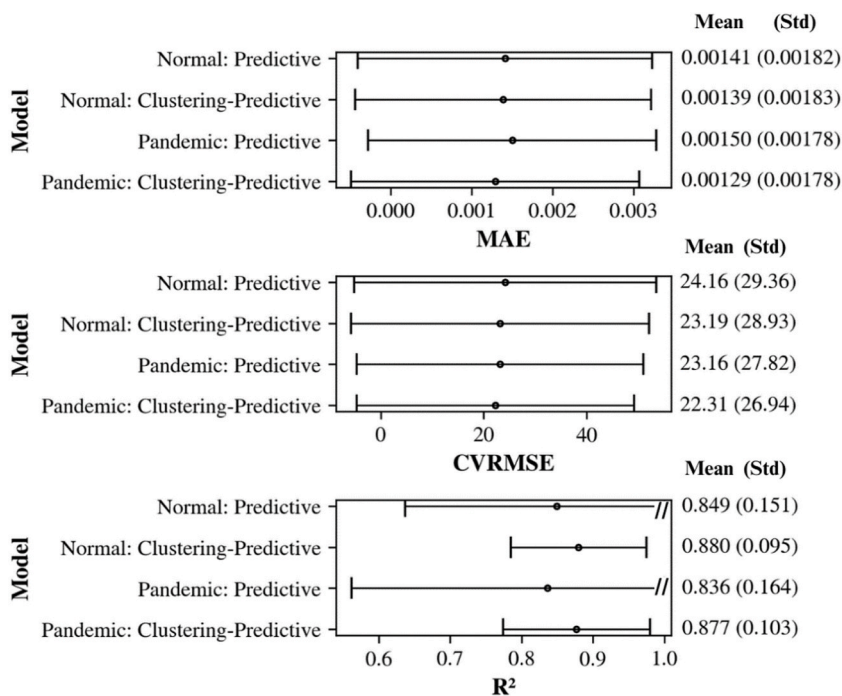


Fig. 13. Performance Metrics Comparison for Predictive-Only and Clustering-Predictive Models Under Normal and Pandemic Conditions: Mean with Standard Deviation as Error Bars. Note: Dots represent the mean values of error metrics, while whiskers extend to one standard deviation, illustrating the distribution and consistency of model performance. Mean plus one standard deviation R^2 values exceeding 1 are truncated to 1 for clarity.

samples, with its standard deviation decreasing by 1.48 %, reflecting greater consistency in performance. The mean R^2 metric increases by 3.65 %, signifying improved explanatory power, while the 55.19 % reduction in its standard deviation highlights more reliable and consistent performance.

During pandemic conditions, the clustering-predictive model exhibits pronounced improvements. The mean MAE decreases by 14.00 %, indicating a significant enhancement in predictive accuracy and effective adaptation to atypical energy consumption patterns, while the standard deviation of MAE remains unchanged, suggesting stable variability. The mean CVRMSE decreases by 3.69 %, indicating sustained predictive reliability under high-variability conditions, with a 3.17 % reduction in its standard deviation pointing to more consistent performance. The mean R^2 metric increases by 4.90 %, demonstrating enhanced explanatory capacity, while the 62.41 % reduction in its standard deviation signifies robust and reliable predictive strength.

These findings highlight the clustering-predictive model's advantages over the predictive-only approach, especially in reducing errors and enhancing stability. Reductions in MAE and CVRMSE, along with lower standard deviations, suggest improved prediction reliability by tailoring forecasts to distinct energy usage patterns within each cluster. The increase in R^2 , combined with a significant decrease in its standard deviation, further confirms the model's strengthened capacity to capture data variability, crucial for robust applications, especially during high-variability periods like the pandemic.

To further examine the effectiveness of the clustering-predictive model, we have illustrated the detailed distribution of its performance indicators in Fig. 14. Under normal conditions, the model demonstrates exceptional predictive accuracy, with over 55.06 % of buildings achieving an R^2 greater than 0.9, and 89.89 % surpassing an R^2 of 0.8. Even during the COVID-19 pandemic, the clustering-predictive model maintains high reliability, with 57.30 % of buildings retaining an R^2 above 0.9 and 85.39 % exceeding 0.8. This enduring performance, irrespective of the prevailing conditions, highlights the model's robustness and establishes its utility as a dependable tool for energy management in both stable and unpredictable environments.

Given the demonstrated superior performance of the clustering-predictive model over the predictive-only approach, subsequent analyses in Sections 3.2.2 and 3.2.3 will focus on examining the spatial and temporal forecasting capabilities of the clustering-predictive model developed in this study.

3.2.2. Performance across building types

This subsection assesses the model's forecasting ability across different building categories to determine how the unique characteristics and data quality associated with each building type influence its predictive accuracy. Fig. 15 visualizes performance metrics for each building type under standard operational conditions, displaying mean values and standard deviations for MAE, CVRMSE, and R^2 . The performance distribution provides insight into both the accuracy and consistency of the model's predictions. It helps in identifying which building types benefit most from the current modeling approach and where further model tuning and data collection efforts should be focused.

For MAE, lower mean values indicate higher predictive accuracy. Research buildings show the best performance with a mean MAE (0.00079), supported by their relatively large sample size (49), which contributes to stable model predictions. This aligns with their controlled energy use, as seen in the yearly EUI distribution, where research buildings cluster tightly around a lower energy use range (Fig. 8(a)). Dormitories also demonstrate good accuracy with a mean MAE (0.00093), aided by their stable occupancy patterns and represented by 10 buildings, contributing to consistent performance. In contrast, office buildings exhibit the highest mean MAE (0.00412), likely due to their smaller sample size (3) and more variable energy usage patterns, as suggested by the wider spread in their yearly EUI. Mixed-use buildings, with a mean MAE (0.00391) and a sample size of 8, face similar challenges due to the complexity of modeling spaces with diverse functions. The standard deviation of MAE indicates prediction consistency; research buildings (0.00045) and dormitories (0.00071) exhibit stable performance, while office buildings (0.00512) and mixed-use buildings (0.00332) show higher variability, reflecting the heterogeneous nature of their energy profiles.

For CVRMSE, which reflects predictive stability, research buildings again perform well with the lowest mean CVRMSE (14.958), indicating reliable predictions. This stability is reinforced by their low standard deviation (16.467), suggesting that the model

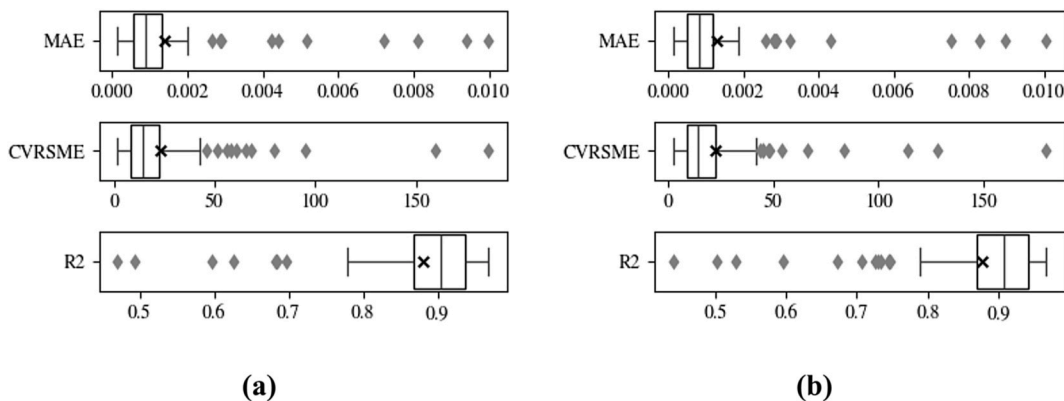


Fig. 14. Boxplots of clustering-predictive performance metrics under (a) normal and (b) pandemic conditions.

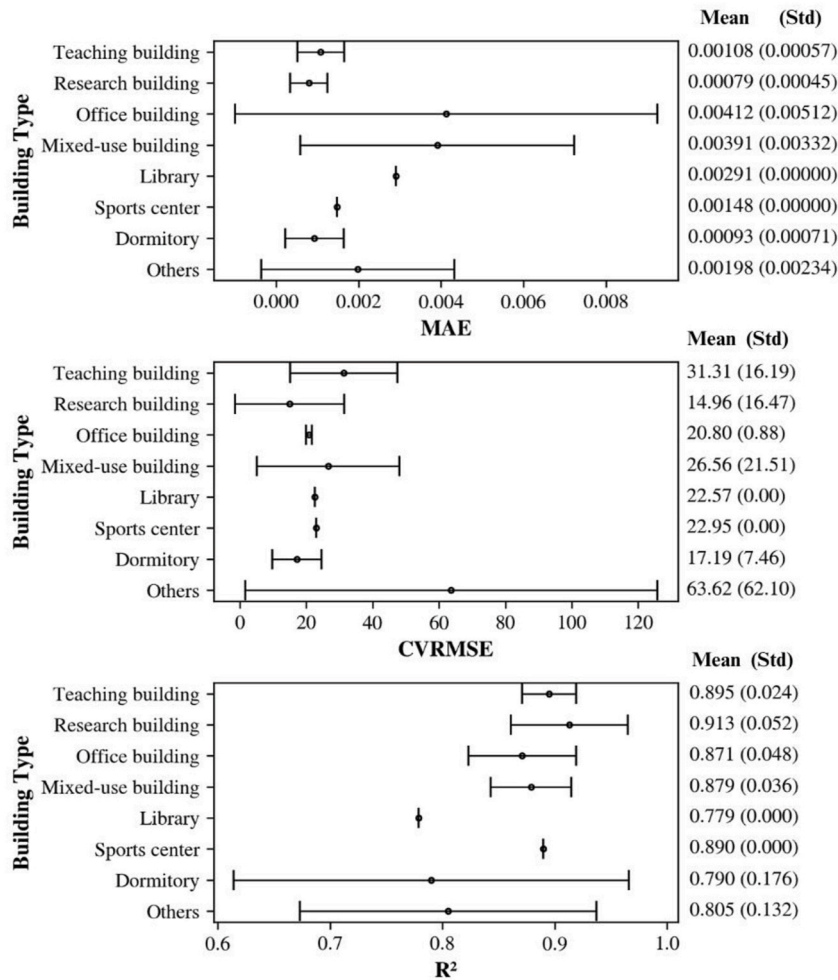


Fig. 15. Performance Metrics Comparison by Building Type: Mean Values with Standard Deviation Error Bars. Note: Dots represent the mean values of error metrics, while whiskers extend to one standard deviation, illustrating the distribution and consistency of model performance.

maintains consistent performance. Dormitories, with a mean CVRMSE (17.188), also show stable predictive performance, supported by predictable energy patterns. Teaching buildings, represented by 7 samples, have a higher mean CVRMSE (31.309), indicating moderate stability but increased variability. The “Others” category, encompassing 10 diverse buildings, has the highest mean CVRMSE (63.622) and standard deviation (62.103), indicating significant challenges in achieving consistent predictions due to varied energy use. The relatively low standard deviation in office buildings (0.875) suggests consistency in predictions, but the higher mean CVRMSE indicates that accuracy remains an issue.

For R^2 , higher mean values indicate stronger explanatory power. Research buildings achieve the highest mean R^2 (0.913), demonstrating the model’s ability to explain a substantial portion of energy use variability, consistent with their concentrated EUI distribution and large sample size. Teaching buildings, with a mean R^2 (0.895), also show strong explanatory power, likely due to their relatively uniform energy patterns. Mixed-use buildings, with a mean R^2 (0.879), display reasonable explanatory power but face challenges due to varied energy profiles, as reflected in their EUI spread. The “Others” category, with a mean R^2 (0.805) and high standard deviation (0.132), reflects limited and inconsistent explanatory power, impacted by the diverse composition of this group. Dormitories, while showing a moderate mean R^2 (0.790), exhibit significant variability (0.176), which suggests that although the model captures energy patterns, there is inconsistency in explanatory power across different samples. This may be due to the fact that dormitories, while generally stable in their energy use, are distributed across multiple clusters in energy pattern analysis, indicating underlying variations that the model struggles to uniformly explain.

Overall, research and teaching buildings show strong model performance with low mean MAE and CVRMSE, high mean R^2 , and low standard deviations, reflecting both accuracy and consistency. Dormitories also perform well in MAE and CVRMSE, though variability in R^2 indicates potential challenges due to underlying differences in energy use patterns, which are evident across clusters despite their functional similarities. This emphasizes the importance of consistent energy patterns and adequate sample sizes for stable model performance. Office and mixed-use buildings present higher errors and variability, likely due to smaller sample sizes and complex

energy profiles. The “Others” category highlights the challenge of modeling heterogeneous building types, shown by high variability across all metrics. Such findings highlight that enhanced data consistency and stability are essential for better model performance, particularly for building types with varied energy use patterns and limited sample sizes.

3.2.3. Performance across epidemic phases

This subsection explores the model's resilience and adaptability by evaluating its performance through various phases of the COVID-19 pandemic, which introduces unique challenges to energy usage forecasting [10]. Fig. 16 provides a visual representation of the model's predictive accuracy across the different epidemic stages, where the one standard deviation interval around the mean R^2 is confined within the 0–1 range to avoid misinterpretation. The analysis across different epidemic periods sheds light on the shifts in model performance over temporal conditions.

Lower mean MAE values indicate better predictive accuracy. In P1, the model achieves the lowest mean MAE (0.00085) and standard deviation (0.00135), suggesting accurate point predictions with minimal fluctuations in performance. This aligns with the initial phase when minimal measures were in place, and energy consumption remained closer to typical levels. P6 also shows good accuracy, with a mean MAE (0.00107) and a moderate standard deviation (0.00150), indicating stable predictions as campus operations normalized. In contrast, P2 and P3 have higher mean MAE values (0.00115 and 0.00134) and standard deviations (0.00168 and 0.00179), reflecting the challenges brought on by the shift to remote teaching and the partial easing of restrictions. These transitions likely introduce fluctuations and irregular spikes in energy usage patterns, impacting point prediction accuracy. The highest mean MAE values are observed in P4 (0.00140) and P5 (0.00153), with larger standard deviations (0.00241 and 0.00222), indicating substantial challenges in precise point prediction accuracy.

When considering CVRMSE, which assesses predictive stability, P4 has the lowest mean (19.854) and a relatively small standard deviation (21.066), highlighting stable and predictable performance under reinforced measures. This consistency reflects the steady energy data in P4 due to stringent preventive measures (Fig. 9), emphasizing that stable data trends contribute to more reliable model performance. P5 also demonstrates stable results with a mean CVRMSE of (20.189) and a standard deviation of (20.043), supporting consistent predictions as activities partially resumed. On the other hand, P2 shows the highest mean CVRMSE (25.506) and standard deviation (41.973), underscoring significant variability and reduced reliability during the shift to remote learning. P1, P3, and P6 exhibit moderate mean CVRMSE values (21.349, 22.662, and 21.479) with higher standard deviations (29.481, 28.438, and 30.130), indicating that small, irregular variations during these phases lead to reduced stability across predictions.

For R^2 , a higher mean implies stronger explanatory power. P3 records the highest mean R^2 (0.874) with the lowest standard deviation (0.109), corresponding to the temporary relaxation of restrictions. During this phase, energy usage becomes more predictable and aligns with typical campus patterns as activities partially resume, making it easier for the model to capture and explain data variability. P6 follows with a mean R^2 (0.855) and the lowest standard deviation (0.093), indicating reliable explanatory power as campus operations approach normalcy. P2, with a mean R^2 (0.841) and a standard deviation (0.142), shows reasonable explanatory

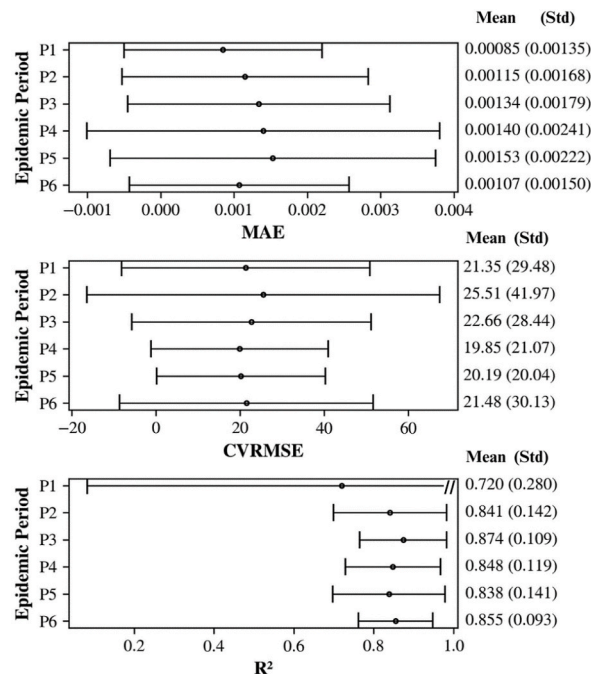


Fig. 16. Mean Values and Standard Deviation Error Bars of Model Performance Across Epidemic Sub-Periods. Note: Dots represent the mean values of error metrics, while whiskers extend to one standard deviation, illustrating the distribution and consistency of model performance. Mean plus one standard deviation R^2 values exceeding 1 are truncated to 1 for clarity.

power but is affected by variability during the transition to remote teaching. P4 and P5, with mean R^2 values of (0.848) and (0.838) and standard deviations of (0.119) and (0.141), demonstrate good explanatory capacity, though with some inconsistencies. P1, characterized by the lowest mean R^2 (0.720) and highest standard deviation (0.637), faces challenges in consistently explaining energy use due to highly unstable and unstructured energy consumption patterns observed during the early pandemic stage (Fig. 9).

In summary, the model's performance during different pandemic phases highlights the impact of energy data stability on predictive accuracy and consistency. In P1, despite a lack of structured energy patterns, the model achieves a low mean MAE, indicating accurate point predictions, though it struggles with explanatory power, as reflected in lower R^2 and higher variability. Phases with stable energy trends, such as P4, show strong performance in predictive stability (low CVRMSE) and explanatory capability (high R^2), supported by steady energy usage under strict preventive measures. Phases like P3 and P6, where activities partially normalize, see higher R^2 values, showing improved explanatory power as energy usage patterns align more closely with typical campus behavior. In contrast, periods like P2, marked by remote teaching shifts, exhibit challenges in stability, impacting CVRMSE and R^2 scores. These findings underscore the importance of stable, structured data in achieving reliable model performance across varied operational conditions. However, the variability in performance across pandemic phases highlights an opportunity for improvement in established forecasting models, particularly in their ability to adapt to significant external disruptions. This observation underscores the necessity for developing adaptive modeling approaches capable of handling the unpredictable and dynamic patterns introduced by unprecedented events, such as the COVID-19 pandemic.

4. Conclusions

This study refines the application of AI-driven clustering-predictive models in electricity demand forecasting by introducing a robust integration of K-means clustering with BiLSTM regression, specifically tailored to address the complexities of university building energy consumption patterns. Unlike previous studies, this research thoroughly evaluates model performance under both normal and pandemic-disrupted conditions, offering insights into its adaptability and precision. The key contributions of this study include: (1) a scalable forecasting framework that eliminates the need for auxiliary variables, enabling large-scale deployment without extensive contextual data; (2) a comprehensive evaluation of the model's performance across various spatial and operational scenarios, demonstrating its robustness in the context of cross-building heterogeneity and under extreme conditions; and (3) insights into the critical role of data stability in achieving reliable load forecasts.

The clustering-predictive model demonstrates significant enhancements in predictive accuracy and stability when compared to a predictive-only approach. Under normal conditions, it exhibits a 1.42 % reduction in mean MAE and a 4.01 % decrease in mean CVRMSE, alongside a 3.65 % increase in R^2 . These results, coupled with a 55.19 % reduction in the standard deviation of R^2 , emphasize the model's strong accuracy and dependability. During pandemic conditions, the improvements are even more pronounced, with a 14.00 % decrease in mean MAE, a 3.69 % drop in CVRMSE, and a 4.90 % increase in R^2 , further strengthened by a 62.41 % reduction in R^2 's standard deviation. These outcomes illustrate the model's robustness in improving forecasting accuracy and consistency, proving essential for maintaining reliable forecasts amidst unpredictable and variable energy use patterns. In addition, the study underscores the importance of data stability and context-specific patterns in achieving reliable energy load forecasts. The model performs best in buildings with consistent energy patterns, such as research and teaching facilities, affirming that stable energy profiles and adequate data volume significantly enhance predictive accuracy and consistency. In contrast, buildings with more complex or heterogeneous energy profiles, like offices and mixed-use spaces, posed greater challenges for the model, revealing limitations in adapting to varied usage patterns when data is limited. The pandemic phases further test the model's adaptability, demonstrating the critical role of data stability in sustaining predictive accuracy. During periods of abrupt operational changes, fluctuations in energy demand impacted performance, underscoring the need for forecasting approaches that can effectively respond to rapid shifts in energy usage. These findings collectively highlight data stability as a cornerstone for achieving reliable performance under both stable and variable conditions.

This research substantially refines the accuracy of short-term electricity load forecasting, delivering vital insights into varied energy usage across multiple spatial and temporal conditions. It underscores the necessity of customizing energy management strategies to align with buildings' distinct dynamics and operational specifics. By advancing beyond mere functional classifications, adopting an approach grounded in energy consumption patterns offers a more comprehensive understanding of the energy demand dynamics in buildings. However, the model's reliance on historical data patterns poses limitations, potentially hampering its adaptability to unprecedented changes. Additionally, while the clustering-based method enhances predictive accuracy, it requires further refinement to effectively manage heterogeneous or sparsely represented building types. Future enhancements should focus on incorporating adaptive strategies that build resilience, allowing the model to dynamically adjust to unpredictable and rapidly evolving conditions. Incorporating auxiliary variables such as weather conditions, occupancy patterns, and operational schedules may also further enhance predictive accuracy, particularly in scenarios with high variability. This work lays a robust foundation for demand management within smart grids and supports advanced applications such as detailed energy profiling and the integration of building systems with renewable energy sources and electric vehicle networks. By pushing the boundaries of nearly zero-energy buildings (NZE) and sustainable urban energy management, this research contributes significantly to the development of more efficient and resilient energy systems.

CRedit authorship contribution statement

Yu-Shin Hu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Kai-**

Yun Lo: Writing – review & editing, Validation, Project administration, Investigation, Formal analysis, Conceptualization. **I-Yun Lisa Hsieh:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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